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A high-accuracy impedance spectrometer for measuring sediments with low polarizability

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Abstract
Spectral impedance measurements are receiving increased attention with regard to the characterization of soils, sediments and rocks, particularly in terms of the internal rock structure, the mineralogical composition and the chemistry of fluids contained in porous rocks. In fluid-saturated, porous sedimentary rocks, which are of particular relevance for many hydrological and environmental problems, the polarization processes that give rise to an observed phase shift between input current and output voltage signals are caused by the interaction of the electrolyte in the pores of the rock with electrically charged mineral surfaces. However, this phase response is relatively weak, typically smaller than 10 mrad and sometimes even of the order of only 1 mrad. In order to reliably measure such phase responses in the relevant frequency range, a high-accuracy impedance spectrometer is required. This system must allow phase measurements between 1 mHz and 1 kHz with a phase accuracy better than 0.1 mrad. In this paper, we present a new impedance spectrometer which meets these requirements. It is based on the four-point measurement method and offers a measurement range from 1 mHz to 45 kHz. Furthermore, we present design information for the sample holder and the electrodes, and methods for performing numerical corrections to reduce measurement errors. The overall accuracy of the setup was validated using water and sand with well-defined polarizable objects.

Keywords: spectral induced polarization, spectroscopy, impedance spectrometer, sediment, soil

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Over the last few years, impedance spectroscopy has received increased attention as a tool for the improved characterization of soils, sediments and rocks in general. Its attractiveness arises from the fact that impedance measurements can actually be performed at the field scale in an imaging framework—an approach that has been shown to provide improved subsurface characterization capabilities for different hydrological and environmental applications (e.g. Kemna *et al* (2000, 2004), Slater and Binley (2006), Hördt *et al* (2007) and Blaschek *et al* (2008)). Originally, impedance spectroscopy—also referred to as the spectral induced polarization (SIP) method in the geophysical community—was mainly applied to study mineralized rocks (Pelton *et al* 1978) where polarization occurs at the interfaces between electronically conducting mineral grains and fluid-filled pores with electrolytic conduction. Over the last decade, the technique has also been increasingly applied to sedimentary rocks without electronically conducting components (e.g. Vanhala (1997), Slater and Lesmes (2002), Scott and Barker (2003), Binley *et al* (2005) and Kemna *et al* (2005)),

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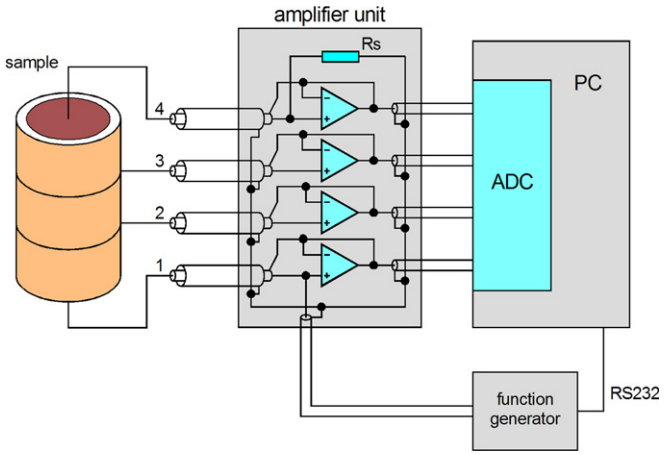


Figure 1. Block diagram of the measurement system with two electrodes (1, 4) for current injection and two electrodes (2, 3) for voltage measurement.

which normally exhibit a much weaker phase response than mineralized rocks. Here, polarization at low frequency (typically <1 kHz) is caused by the interaction of the pore fluid (electrolyte) with electrically charged mineral surfaces, where the so-called electric double layer is formed (see, e.g., Leroy *et al* (2008)). SIP properties of sediments are connected with state variables, such as pore fluid salinity (Lesmes and Frye 2001, Kemna *et al* 2005) and saturation (Ulrich and Slater 2004, Binley *et al* 2005), as well as structural and mineralogical characteristics, such as pore size (Scott and Barker 2003, Binley *et al* 2005) and clay content (Slater and Lesmes 2002), which are relevant for many hydrological and environmental issues. However, the phase response of sediments is typically smaller than 10 mrad and sometimes even of the order of only 1 mrad. In order to reliably measure such phase responses in the laboratory, a special measurement system with sample holder is required. Available impedance spectrometers were mainly developed for electrochemical or material analysis (Macdonald 1987) but special methods and instruments have also been developed for geoscientific applications (e.g. Emerson (1969) and Vanhala and Soininen (1995)). The goal of this study was to develop a system with improved phase accuracy. Based on these methods, a new impedance spectrometer and measurement technique will be presented in this paper that is capable of measuring the spectral induced polarization response of sediments in the range from 1 mHz to 45 kHz with a phase accuracy better than 0.1 mrad up to 1 kHz. Relevant measurement errors will be analyzed in order to assess the accuracy, to derive methods for numerical correction and to design the sample holder and the electrodes. Our measurement technique allows the accurate measurement of the spectral induced polarization response of sediment samples.

2. Measurement system

Our measurement system (see figure 1) uses the four-point measurement method to determine the impedance across a sample (assumed to be homogeneous). It comprises a function generator, an amplifier unit, an ADC card, a PC and a sample holder with two electrodes for current injection and two electrodes for voltage measurement.

The function generator (Agilent 33120A) generates a sinusoidal excitation voltage with stable frequency in the range from 1 mHz to 45 kHz (typically with a peak-to-peak amplitude of 10 V). Amplitude and frequency of the signal are remote controlled.

The amplifier unit measures the voltages at the sample electrodes with minimum load and cross talk. The voltages at the electrodes are transmitted over short triaxial cables (0.5 m) to the high-impedance low-capacity dc-coupled amplifiers. The triaxial cables are used with a driven shield (Morrison 1998) to reduce parasitic leakage currents and capacitive loads. The input capacity, measured between the inputs of the triaxial cables and ground, is less than 5 pF and the input resistance is much greater than 1 GΩ. The outer shield is connected to ground potential to minimize cross talk. The coupling capacities of the amplifiers are also minimized to reduce cross talk. They are smaller than 0.1 pF. Several operational amplifiers (OP) were tested and the commercial JFET OP AD825 was chosen for the four amplifiers in the device because of its high input impedance and stability. The slew rate is 140 V μs^{-1} and the input resistance is 500 GΩ. A shunt resistor R_s (10 Ω, 100 Ω or 1 kΩ) is used to measure the current. All channels are constructed in an identical manner in order to minimize phase and gain differences between the channels.

The data acquisition card NI4472 simultaneously digitizes the voltages with 24-bit Sigma-Delta ADCs with high phase accuracy and with digital anti-aliasing filters. The oversampling rate is 64 at the maximum sampling rate of 102 kHz, and the maximum measurement frequency is ≈45 kHz. The voltage range of ±10 V is large enough to overcome the redox potential at the electrodes, and the resolution of 24-bit (1.2 μV) enables the ac voltages to be measured with an acceptable signal-to-noise ratio. The cross talk of the DAQ card is smaller than −90 dB (shorted input).

LabVIEW and MATLAB programs were developed for data acquisition and post-processing. The LabVIEW program controls the function generator and the ADC card. For each measurement frequency, the time series of the voltages are stored. The MATLAB program removes drifts in the time series as proposed by Dahlin *et al* (2002) and computes the complex voltages $U(f)$ (phasors) from the measured time series $u(t)$ at the known excitation frequencies $f = 1/T$,

$$U(f) = \frac{2}{T}\int_0^T u(t)\cos(2\pi f t)\,\mathrm{d}t - \mathrm{j}\frac{2}{T}\int_0^T u(t)\sin(2\pi f t)\,\mathrm{d}t, \quad (1)$$

the impedance Z_x between electrodes 2 and 3 (see section 3.3) and the impedance Z_{14} between electrodes 1 and 4. Here, T and j are, respectively, the period of the signal and the imaginary unit. In addition to the statistical measurement uncertainty of the impedances, the noise and distortion factors are estimated.

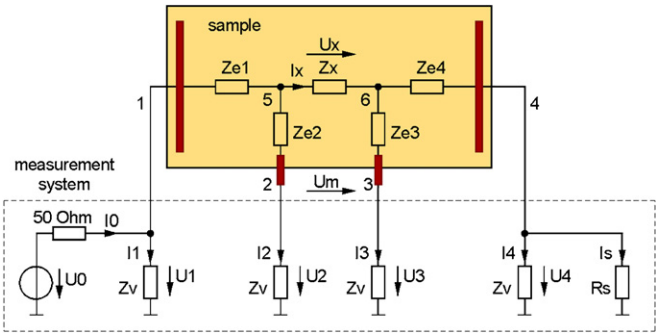


Figure 2. Simplified electrical model of the sample and the measurement system.

3. Improved impedance determination and error estimation based on an electrical model

The discrepancies between the measured values and the true values of the sample are estimated by means of an electrical model which represents the sample and the measurement system (see figure 2). Let Z_x denote the impedance across the sample, Z_{e1} and Z_{e4} the contact impedances at the current electrodes including the remaining part of the sample and Z_{e2} and Z_{e3} the contact impedances at the potential electrodes. The model also comprises the input impedances Z_v of the amplifiers (including the cables), the shunt resistor R_s and the voltage source U_0. Since the system was designed for ground-based voltage measurements with minimal cross talk between the electrodes, no additional coupling impedances between the electrode cables must be considered when describing the electrical properties of the measurement system.

In order to calculate the true sample impedance $Z_x = U_x/I_x$, the true voltage U_x and the true current I_x must be known. However, only the voltages and currents outside the sample can be measured. Hence, in the following, the discrepancy between U_x and U_m as well as the discrepancy between I_x and I_s are discussed and methods for more accurate calculations of U_x and I_x are proposed. The equations in the following section were validated with nodal analyses of the electrical model.

3.1. Calculation of the true excitation current I_x

First, the discrepancy between the currents I_x and I_s is estimated. Figure 2 shows that the current I_s through the shunt resistor is smaller than the current I_x through the sample. This is because of the parasitic currents I_3 and I_4. By simple analysis of the linear electrical network, the discrepancy between the measured current I_s and the true current I_x can be calculated as

$$\frac{I_s}{I_x} = \frac{Z_v + Z_{e3}}{Z_v + Z_{e3} + 2R_s + Z_{e4} + \frac{Z_{e3}R_s + Z_{e4}R_s}{Z_v}}. \tag{2}$$

Equation (2) can be simplified by taking into account that $Z_v \gg Z_{e3}$, $Z_v \gg Z_{e4}$ and $Z_v \gg R_s$, which yields

$$\frac{I_s}{I_x} \approx \frac{1}{1 + (2R_s + Z_{e4})/Z_v}. \tag{3}$$

From equation (3), it is obvious that the discrepancy between I_s and I_x can be decreased by using a small shunt resistor R_s; however, it cannot be removed completely, since it also depends on Z_{e4}. A different system design, which would measure the current I_0 with a shunt resistor at electrode 1, would not be useful. Due to the higher voltage at electrode 1 in relation to the voltage at electrode 4, the leakage currents I_1 and I_2 are higher than the leakage currents I_3 and I_4. The discrepancy between I_0 and I_x is thus larger than the discrepancy between I_s and I_x. Therefore, the best option is to measure the current using a shunt resistor between electrode 4 and ground.

A more accurate current measurement can be achieved by calculating the true value of I_x. This is possible because of the ground-based voltage measurement of U_3 and U_4 and the well-known impedances Z_v. The true excitation current I_x is the sum of current $I_s = U_4/R_s$ and the leakage currents $I_3 = U_3/Z_v$ and $I_4 = U_4/Z_v$. This leads to

$$I_x = I_s + I_3 + I_4 = \frac{U_4}{R_s} + \frac{U_3}{Z_v} + \frac{U_4}{Z_v}. \tag{4}$$

The impedances Z_v are primarily determined by the input capacities of the amplifiers, which are about 5 pF. However, in order to accurately calculate the current I_x, the impedances Z_v are measured.

3.2. Accuracy of the voltage measurement

The discrepancy between the true voltage U_x and the measured voltage $U_m = U_2 - U_3$ (the voltage between electrodes 2 and 3) is caused by the load of the amplifier Z_v and the impedances Z_{e2} and Z_{e3}. The impedances form the voltage dividers D_2 and D_3:

$$D_2 = \frac{U_2}{U_5} = \frac{1}{1 + Z_{e2}/Z_v}, \qquad D_3 = \frac{U_3}{U_6} = \frac{1}{1 + Z_{e3}/Z_v}, \tag{5}$$

which results in the discrepancy between U_x and U_m:

$$\frac{U_m}{U_x} = \frac{U_2 - U_3}{U_5 - U_6} = \frac{U_5 D_2 - U_6 D_3}{U_5 - U_6}. \tag{6}$$

For equal impedances $Z_{e2} = Z_{e3} = Z_{e2,e3}$, respectively equal voltage dividers $D_2 = D_3 = D$, equation (6) simplifies to

$$\frac{U_m}{U_x} = D = \frac{1}{1 + Z_{e2,e3}/Z_v}. \tag{7}$$

From equation (7), it is obvious that the discrepancy between U_m and U_x can be decreased by using a good contact between the voltage electrodes and the sample so that the impedances $Z_{e2,e3}$ are small.

The error in the voltage measurement can be avoided if the impedances Z_{e2} and Z_{e3} are taken into account. In this case, the true voltage $U_x = U_5 - U_6$ can be computed according to

$$U_x = U_2\left(1 + \frac{Z_{e2}}{Z_v}\right) - U_3\left(1 + \frac{Z_{e3}}{Z_v}\right). \tag{8}$$

However, for real samples, the impedances Z_{e2} and Z_{e3} are often unstable and unknown. In order to apply this correction, they should be estimated during the impedance measurement.

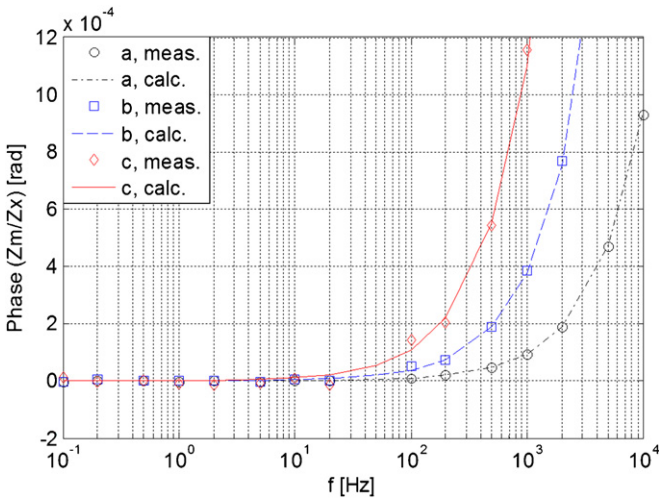


Figure 3. Measured and calculated phase responses of a resistor network with varying Z_{e4} (a: 1 kΩ, b: 10 kΩ and c: 33 kΩ). The network corresponds to the electrical model of the sample in figure 2 with $Z_x = 1$ kΩ, $R_s = 1$ kΩ and $Z_{e2} = Z_{e3} = 0$ Ω.

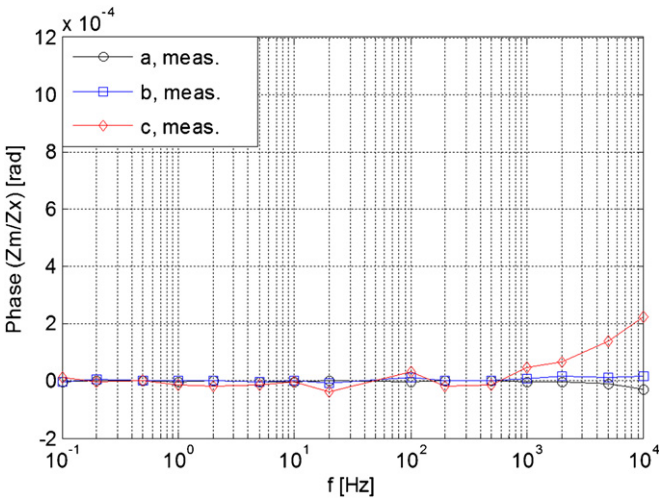


Figure 4. Measured phase response of a resistor network with current correction with varying Z_{e4} (a: 1 kΩ, b: 10 kΩ and c: 33 kΩ). The network corresponds to the electrical model of the sample in figure 2 with $Z_x = 1$ kΩ, $R_s = 1$ kΩ and $Z_{e2} = Z_{e3} = 0$ Ω.

3.3. Accuracy of the impedance measurement

Since the impedances Z_{e2} and Z_{e3} are unknown, usually only the current correction (4) is used for impedance calculation. In this case, the error of the impedance measurement—the discrepancy between the measured value $Z_m = U_m/I_x$ and the true value $Z_x = U_x/I_x$—can be determined as

$$\frac{Z_m}{Z_x} = \frac{1}{1 + Z_{e2,e3}/Z_v}. \tag{9}$$

The error of the impedance measurement without using corrections (4) and (8) can be estimated as

$$\frac{Z_m}{Z_x} = \frac{U_m}{U_x}\frac{I_x}{I_s} \approx \frac{1 + (2R_s + Z_{e4})/Z_v}{1 + Z_{e2,e3}/Z_v}. \tag{10}$$

3.4. Estimation of tolerable impedance values

The phase responses of a well-defined resistor network, which corresponds to the electrical model of the sample in figure 2, are measured and calculated to estimate tolerable values for the impedances Z_e and to validate equations (4), (9) and (10). To begin with, the phase response was measured and calculated using (10) for values $Z_{e4} = 1$ kΩ, 10 kΩ and 33 kΩ. The other impedances were $Z_x = 1$ kΩ, $R_s = 1$ kΩ and $Z_{e2} = Z_{e3} = 0$ Ω. The input impedances Z_v were simplified as a parallel circuit with an ohmic conductance $G_v = 1$ nS and a capacity $C_v = 5$ pF.

The measured phase responses of Z_m/Z_x in figure 3 confirm the theoretical estimations and they show a positive phase. Due to the capacitive load of the amplifiers, the phase responses increase with frequency. This phase error can be decreased using the current correction (4). The phase errors after correction are smaller than 0.1 mrad for frequencies below 3 kHz (see figure 4).

These calculations and measurements indicate the maximal tolerable values for Z_{e4} and R_s. Without current correction, the resistance $2R_s + Z_{e4}$ (according to equation (10) and figure 3) should be smaller than approximately 3 kΩ for a phase error smaller than 0.1 mrad at a frequency $f = 1$ kHz. With current correction, the resistance is approximately 35 kΩ (according to figure 4) for the same degree of accuracy.

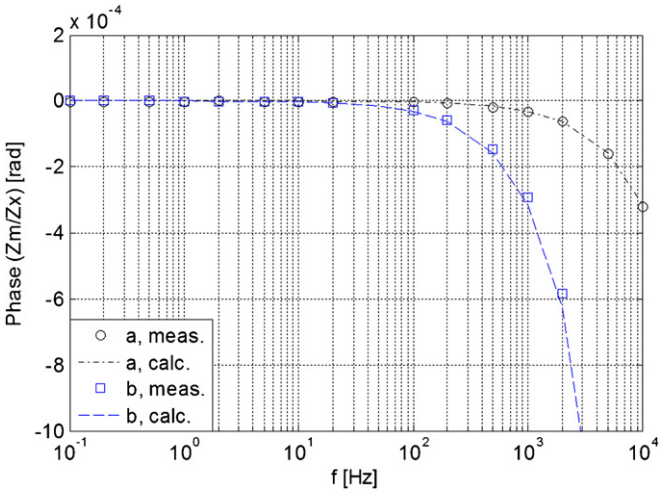


Figure 5. Measured (with current correction) and calculated phase response of a resistor network with varying $Z_{e2,e3}$ (a: 1 kΩ and b: 10 kΩ). The network corresponds to the electrical model of the sample in figure 2 with $Z_x = 1$ kΩ, $R_s = 1$ kΩ and $Z_{e1} = Z_{e4} = 0$ Ω.

In the second stage, the phase response was measured with current correction and calculated using equation (9) for values $Z_{e2,e3} = 1$ kΩ and 10 kΩ. The other impedances were $Z_x = 1$ kΩ, $R_s = 1$ kΩ and $Z_{e1} = Z_{e4} = 0$ Ω. The measured phase responses of Z_m/Z_x in figure 5 also confirm the theoretical estimations, and they show a negative phase response. These measurements and estimations indicate the maximum tolerable value for Z_{e2} and Z_{e3}. This value should be smaller than 3 kΩ for a phase error smaller than 0.1 mrad at a frequency of $f = 1$ kHz.

3.5. Common mode error

The common mode voltage $U_c = (U_5 + U_6)/2$ (Tietze and Schenk 1985) can also cause measurement errors because the voltages U_5 and U_6 are usually measured with different gains G_2 and G_3 of the channels and different voltage dividers D_2 and D_3 (see equation (5)). The discrepancy between the measured voltage after digitizing U_a and U_x is given as (see also equation (6))

$$\frac{U_a}{U_x} = \frac{G_2U_2 - G_3U_3}{U_5 - U_6} = \frac{G_2D_2U_5 - G_3D_3U_6}{U_5 - U_6}. \quad (11)$$

For a simple estimation of the common mode error in the lower frequency range, the effect of different impedances Z_{e2} and Z_{e3}, respectively different voltage dividers D_2 and D_3, can be neglected. The impedances Z_v are mainly affected by the capacitive load of the amplifiers, and therefore the impedances are very high at low frequencies. This results in voltage dividers D_2 and D_3 which are equal to 1. The voltages U_5 and U_6 can be calculated as $U_5 = U_6 + I_xZ_x$ and $U_6 = I_x(Z_{e4} + R_s)$ and the measured current after digitization can be calculated as $I_a = G_4I_x$ (G_4 is the gain of the voltage U_4). The discrepancy between the measured impedance Z_a and the true impedance Z_x therefore results in

$$\frac{Z_a}{Z_x} = \frac{U_a}{U_x}\frac{I_x}{I_a} = 1 + \frac{G_2 - G_4}{G_4} + \frac{G_2 - G_3}{G_4}\frac{Z_{e4} + R_s}{Z_x}. \quad (12)$$

Here, the second term $(G_2 - G_4)/G_4$ on the right-hand side causes a constant error that is independent of the impedance values. However, the third term $(Z_{e4} + R_s)/Z_x$ is more problematic. It depends on the impedance Z_{e4}. The current electrodes used normally cause large polarization effects in the lower frequency range. The phase of Z_{e4} (and Z_{e1}) is of the order of some hundred mrad (see section 4). Part of the impedance can be measured if the gains G_2 and G_3 are different. For instance, a gain difference $(G_2 - G_3)/G_4$ of 0.1% and an impedance $Z_{e4} = Z_x \exp(0.5j)$ with a phase difference of 500 mrad between Z_{e4} and Z_x cause a phase error of about 0.5 mrad. This estimation shows that a higher common mode rejection and a small impedance Z_{e4} (and a small R_s) in relation to Z_x are necessary if this error is to be avoided.

The common mode error caused by small differences in the transfer functions of the channels can be minimized by an additional numerical correction of the measured complex voltages $U_1(f) \cdots U_4(f)$. Therefore, the gain functions $G_1(f) \cdots G_4(f)$ of the channels are measured with the same input signal. The measured gain differences are up to 0.1% for frequencies below 10 kHz. After the numerical correction has been applied, the differences are smaller than 0.002% for the complete frequency range. This method can minimize the error caused by different gains, but it cannot minimize the phase error in the high-frequency range as a result of different voltage dividers.

To estimate the phase error in the high-frequency range due to different impedances Z_{e2} and Z_{e3}, respectively different voltage dividers D_2 and D_3, the voltage U_5 in equation (6) is expressed as kU_6 with some factor k, resulting in

$$\frac{U_m}{U_x} = D_2 + \frac{D_2 - D_3}{k - 1}. \quad (13)$$

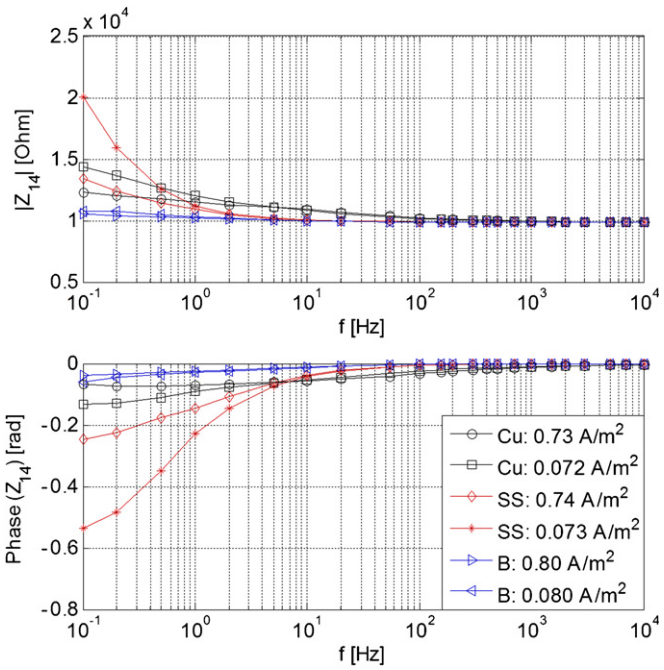


Figure 6. Measured impedance Z_{14} for different current densities and electrode materials (copper (Cu), stainless steel (SS), porous bronze (B)).

The phase is now calculated for $k = 2$ where the difference voltage $U_x = U_5 - U_6$ is equal to the voltage U_6. This is the case corresponding to the preferred sample holder design described in section 7. The phases for the two extreme cases $Z_{e2} = 0\ \Omega$, $Z_{e3} = 3\ \text{k}\Omega$ and $Z_{e2} = 3\ \text{k}\Omega$, $Z_{e3} = 0\ \Omega$ are about +0.1 mrad and −0.2 mrad, respectively, for the maximum tolerable impedance values of $Z_{e2,e3} = 3\ \text{k}\Omega$ (see section 3.4). This means that the phase error for the second extreme case is 0.2 mrad, and that the phase can change from positive to negative values if the impedances are different.

4. Contact impedance of the electrodes

In order to choose a suitable material for the current electrodes, the impedances Z_{14} between two plates (here, made of copper, stainless steel or porous bronze) were measured using a water-filled sample holder (tap water resistivity = 21 Ω m, width = 6 cm, height = 1 cm, length = 28 cm) for two different current densities. The bronze plates are filter elements with an average pore diameter of 15 μm, a thickness of 4 mm and a porosity of 32%; the material is Cu89Sn11. Prior to their use the copper and stainless steel plates were polished, while the porous bronze plates were used without any surface treatment. After the sample holder had been prepared, we waited several hours before starting the measurements. The results are shown in figure 6.

At frequencies above 10 Hz, the impedance Z_{14} shows little variation with frequency and is almost independent of the electrode material and current density. The impedance depends only on the sample material and on the shape of the sample holder. At lower frequencies, the response shows a spectral variation as a result of the electrode effects. For the

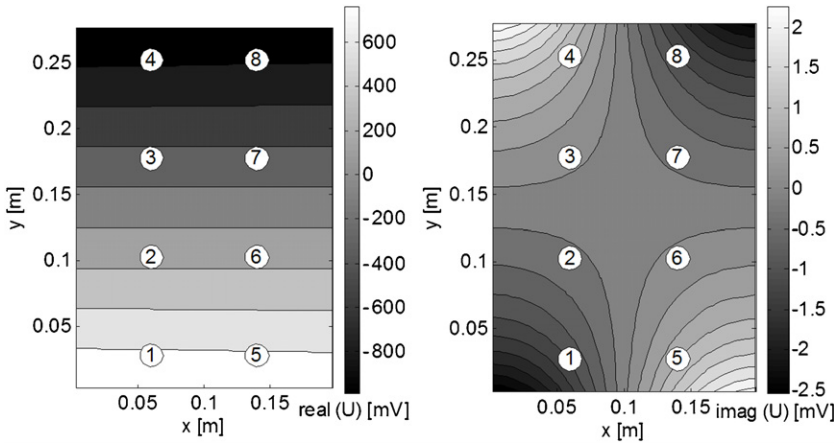


Figure 7. Calculated potential distribution of a flat sample with location-dependent contact impedances. Left: real component. right: imaginary component.

selected materials, the lowest phase shifts and impedances are obtained with the porous bronze electrodes.

The impedance is smaller for higher current densities. In addition, the high stimulus voltage improves the signal-to-noise ratio. Therefore, we conclude that it is useful to conduct measurements with high current densities. The presented system typically uses an excitation voltage of ±5 V with current densities in the range of 0.01–2 A m^{-2}.

We did not find significant nonlinear effects. Typically, the contact impedance shows nonlinear effects for a current density larger than 10^{-3} A m^{-2} (Sumner 1976). Due to the nonlinear impedances, the voltage source used can cause excitation currents with harmonics. However, several tests have shown that these harmonics have no influence on the accuracy of the measurement. This is because only the basic oscillation is used for the impedance calculation of Z_x. Moreover, Vanhala and Soininen (1995) showed in their studies that the observed IP effect of the sample does not change for current densities varying from 10^{-5} A m^{-2} to 0.2 A m^{-2}.

It shall be noted that the inverse relationship between current density and contact impedance can cause an inhomogeneous current distribution on the surface of the current electrodes. Areas of the electrodes with low impedance will exhibit a high current density, which in turn lowers the impedance even more. This positive feedback makes an inhomogeneous current distribution likely.

5. Phase error due to inhomogeneous contact impedances

Inhomogeneous contact impedances of the current electrodes can cause another phase error. We investigated this error for sample holders of cubical and cylindrical shape containing homogeneous samples. In order to establish homogeneous current distribution, low contact resistance metal plates (see above) were mounted at the ends of the sample holder. The polarizability of these current electrodes is relatively large and the contact impedance can be location-dependent, because of oxidation or other effects (see section 4). To analyze these effects, we first computed the 2D potential distribution of a flat rectangular tank filled with water (tap water resistivity $\rho \approx 21\ \Omega$ m, length $l_y = 0.28$ m, width $l_x = 0.20$ m, height $h = 0.01$ m) assuming current electrodes (width 0.20 m) with inhomogeneous contact impedances. We simulated electrodes with changing contact impedance in the x direction, $\partial Z_x/\partial_x$, which increased linearly from 1953 exp(−0.28j) Ω m^{-1} to 2387 exp(−0.28j) Ω m^{-1}.

Ideally, the partial derivative of the potential in the x direction, $\partial U/\partial x$, should be zero. However, the computed distribution of the imaginary component of the potential (see figure 7) shows a dependence on x. This means that the result of an impedance measurement depends on the position of the potential electrodes. For example, the potential difference between positions 1 and 4 is −1.8 mV with a negative phase and the potential difference between positions 5 and 8 is +1.8 mV with a positive phase. Furthermore, the computation shows that the greater the distance between a potential electrode and a current electrode, the smaller is the phase error. For instance, the calculated phase of the potential difference between positions 1 and 2 is −1.3 mrad and the phase of the potential difference between positions 2 and 3 is −0.8 mrad. Real measurements, which were done on the water-filled 2D tank, showed the same behavior.

The required minimum distance between a current and a potential electrode of the sample holder was then estimated in order to achieve an accuracy of 0.1 mrad. For this, the phase error was measured in relation to the distance x between the current and potential electrodes (6 cm and 12 cm) for different materials of the current electrodes (stainless steel and porous bronze) in a water-filled sample holder with a rectangular cross section of 5.8 cm by 5.0 cm (see figure 8). The distance between the potential electrodes, y, was 12 cm.

The measured phase responses at low frequencies were much lower for $x = 12$ cm than for $x = 6$ cm (see figure 9) and the phase error ($x = 12$ cm) was less than 0.1 mrad for porous bronze and stainless steel.

Therefore, we deduce that the ratio of the distance between the current and potential electrodes and the width of the sample

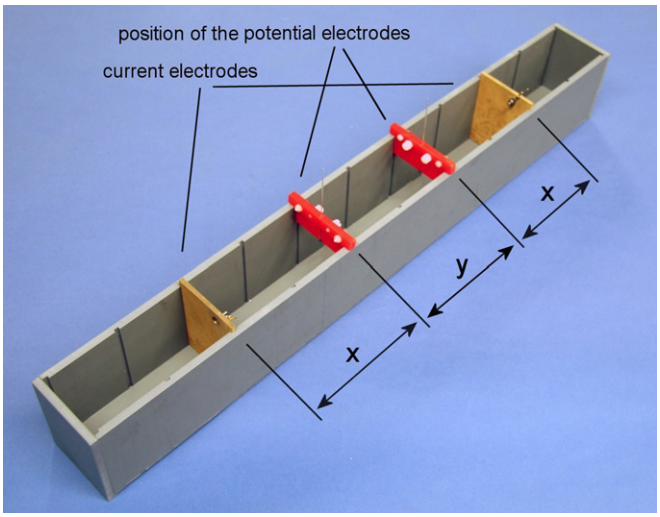


Figure 8. Design of the sample holder used for test measurements.

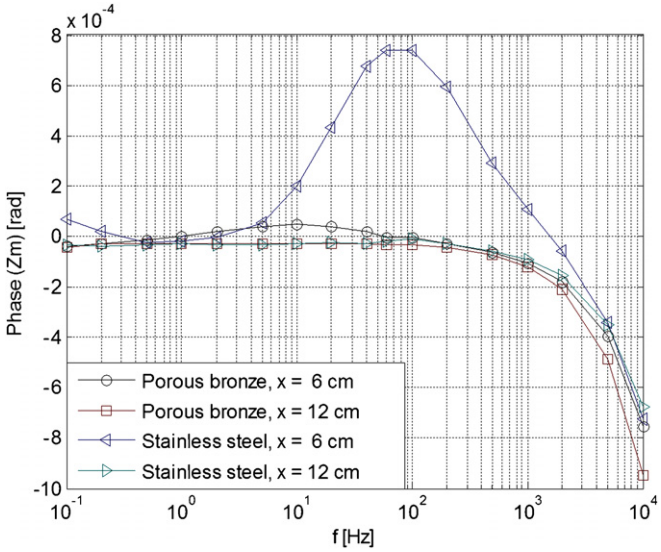


Figure 9. Phase spectra of Z_m measured for different distances between current and potential electrodes (6 cm and 12 cm) for two types of current electrodes (stainless steel and porous bronze).

should be about 2:1 or more. In order to reduce the phase error further, the contact impedance must be small in relation to the sample impedance. A porous material with a large contact area is therefore preferable for the current electrodes.

6. Polarization effects of the potential electrodes

Typically, metallic material is also used for the potential electrodes. The electrical polarization at the surface of an electronic conductor is caused by the electric current flowing across the interface between the electrolyte and the electronic conductor (see, e.g., Eskola (1992) and Guptasarma (1983)). In order to minimize this polarization, the electric field produced by the excitation current should be zero at the electrode. This can be achieved by placing the metallic electrodes outside the sample and by using an electrolyte for

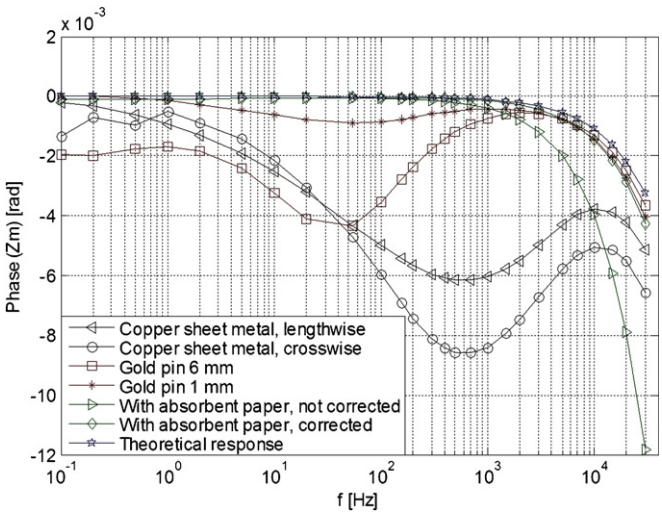


Figure 10. Measured phase spectra for different types of potential electrodes and theoretical phase spectrum of water (21 Ω m, $\varepsilon_r = 80$).

the electrical connection between the sample and the metal. This approach is used to avoid polarization of the electrodes.

To demonstrate these polarization effects, the phase response was measured using potential electrodes of different sizes and types and a sample holder filled with water (length = 28 cm, width = 6 cm, height = 1 cm, tap water resistivity $\rho \approx 21$ Ω m).

The impedance of water measured with copper sheet electrodes (width 11 mm) shows phases of −6 mrad and −9 mrad at 600 Hz (see figure 10). The smaller gold electrode (diameter 6 mm) shows a phase of −4 mrad at 40 Hz and the smallest gold electrode (diameter 1 mm) only −1 mrad at 50 Hz. The smallest values were obtained with the external electrodes, which use water or a water-saturated carrier, such as absorbent paper or porous ceramic, for the electrical connection between the sample and the metal. Unfortunately, the external electrode with the absorbent paper (diameter 8 mm) caused an error in the high-frequency range because of its high contact impedance. This error was numerically corrected (see equation (8)) by assuming a contact resistance of 8 kΩ. However, it is better to use electrodes with a larger cross-sectional area and a higher conductivity, such as the bigger porous ceramic electrode used for the measurements in figure 11 (see section 7). This construction is comparable to classic non-polarizing electrodes which use a ceramic membrane between the sample electrolyte and the electrolyte of the electrode. Such a design allows using an electrolyte with a higher conductivity around the electrodes to decrease the contact resistance.

7. Sample holder design

Based on the considerations in the previous sections, we designed a sample holder that allows a phase accuracy of 0.1 mrad in the frequency range from 1 mHz to 1 kHz. The cross section of the sample holder is rectangular or circular, and the current electrodes are porous bronze plates mounted

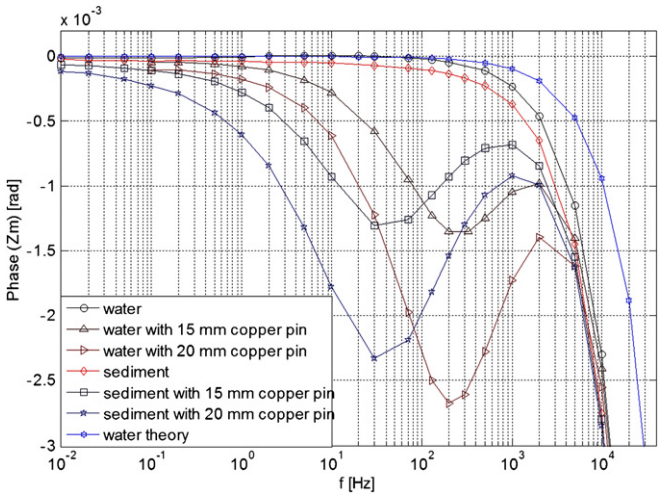


Figure 11. Measured phase spectra of water (21 Ω m) and saturated silica sand (grain size range: 0.1–0.5 mm) with and without copper pins (length 15 mm and 20 mm, respectively, diameter 1.7 mm) placed in the center of the sample holder (orientation in the direction of current flow) and theoretical phase spectrum of water (21 Ω m, $\varepsilon_r = 80$).

at the ends. It is assumed that the entire sample holder is filled with one homogeneous material.

In order to avoid common mode errors, Z_{e4} should not be larger than Z_x (see section 3.5). Hence, the distance between the potential electrodes should be equal to or greater than the distance between the potential electrodes and the current electrodes. Therefore, the potential electrodes are fixed symmetrically at 33% and 66% of the length of the sample holder, so that $Z_{e1} \approx Z_x \approx Z_{e4}$. Errors associated with the polarization of the current electrodes (see section 5) can be avoided if the distance between the current electrodes and the potential electrodes is twice the sample width or more. For the width, one-sixth of the distance between the current electrodes is chosen. Furthermore, the contact resistance of the current electrode should be small in relation to the sample resistance. Therefore, a porous metal such as porous bronze is used as the current electrode material.

The size of the sample holder is determined by the maximum rated value of the impedance Z_{e4} (see section 3.4), which should be less than 35 kΩ, and the resistivity of the sample material. A design for samples with a maximum resistivity of 1000 Ω m results in a sample holder with a cross section of 6 × 6 cm^2 and a length of 12 cm between the voltage electrode 3 and the current electrode 4 (see figure 2) when the previous relation between the length and width of the sample holder is used so that the resistance of Z_{e4} is smaller than 35 kΩ. The total length of the sample holder is 36 cm.

For the potential electrodes, hollow porous ceramic cones filled with water are used to avoid polarization effects and to obtain low contact impedances Z_{e2} and Z_{e3}. The mean diameter of the area in contact with the sample is 13 mm and the contact length is 30 mm.

In order to avoid errors caused by parasitic capacities, the sample should be far away from any ground potential or conducting materials.

8. Test measurements

We performed measurements on a water sample, which has a known phase spectrum, and on test samples with low phase responses (see figure 11) to verify the accuracy and the resolution of the system and the sample holder. The phase error of the water measurement (tap water, resistivity about 21 Ω m, assumed permittivity $\varepsilon_r = 80$) is much smaller than 0.1 mrad for frequencies up to 100 Hz and only about 0.1 mrad at 1 kHz. No voltage correction was used to compensate for the effect caused by the contact impedances of the potential electrodes.

To get well-defined small phase responses, thin copper pins (length 15 mm and 20 mm, respectively, diameter 1.7 mm) were placed at the center of the sample holder filled with water (orientation in the direction of current flow). The measured phase shifts in the medium frequency range, at about 250 Hz, were only −1.3 mrad (15 mm pin) and −2.7 mrad (20 mm pin).

The same measurements were repeated with saturated silica sand (saturated with the same tap water, grain size of the silica sand ranging from 0.1 mm to 0.5 mm). The phase response of the sediment without a copper pin was similar to the pure water measurement. The difference between the phase values was very small, 0.1 mrad at 100 Hz. The phase values of the measurements with copper pins in sand were also similar to the measurements without sand: −1.3 mrad and −2.3 mrad, respectively, for the two copper pins (15 mm and 20 mm) at the respective local minimum in the medium frequency range. However, the frequency of the local minima was shifted from 250 Hz to about 140 Hz.

These results verify the high accuracy and high resolution of the system, which allows for discriminating samples with very small phase responses. It should also be noted that further measurements were performed with this system. For example, Kemna *et al* (2005) conducted measurements to investigate the relationship between the electrical and hydraulic properties of sands.

9. Summary

In this paper, we presented an impedance spectrometer with high phase accuracy and high resolution based on the four-point measuring method. The system is capable of measuring the spectral induced polarization of sediments with an accuracy better than 0.1 mrad in the frequency range from 1 mHz to 1 kHz. The electrodes are connected with short triaxial cables to high-resistance FET amplifiers. The driven shield method is used to reduce the capacitive load to values less than 5 pF. The cross talk between the electrode signals is reduced by means of an optimized layout of the amplifier and the grounding of the outer shield of the triaxial cables. The sinusoidal stimulus voltage, typically 5 V, is generated with a crystal-controlled function generator. For data acquisition, a high-resolution ADC card with 24-bit, 8 channels, simultaneous sampling, a sampling rate of 102 kHz at an oversampling rate of 64, digital anti-aliasing filters and a voltage range of ±10 V is used. A LabVIEW program controls the system and measures

the time series. A MATLAB program calculates the complex amplitudes, the nonlinear distortion and noise, the impedances of the sample between electrodes 2–3 and 1–4 and corrects for errors in the measurements.

The most important measurement errors were investigated to optimize a sample holder with a rectangular or circular cross section and to improve the numerical calculation of the impedances. The current electrodes of the sample holder are porous bronze plates mounted at the ends. The width of the sample holder is one-sixth of the distance between the current electrodes in order to minimize polarization errors. The length of the sample holder for sediments with a maximum resistivity of 1000 Ω m is 36 cm. The total impedance is smaller than 100 kΩ for this size. We suggest the use of hollow porous ceramic cones filled with water as non-metallic potential electrodes. These should be positioned symmetrically at 33% and 66% of the length of the sample holder. In addition, some measurement errors can be reduced by using numerical corrections, such as the error of current and voltage measurements and the common mode error. In order to reduce the contact resistance of the electrodes and to increase the signal-to-noise ratio, the system uses a high stimulus voltage and a high current density.

Measurements on water with well-defined polarizable objects demonstrated the high accuracy of the system, allowing for the discrimination of samples with very small phase responses.

References

Binley A, Slater L D, Fukes M and Cassiani G 2005 Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone *Water Resources Res.* **41** W12417

Blaschek R, Hördt A and Kemna A 2008 A new sensitivity-controlled focusing regularization scheme for the inversion of induced polarization data based on the minimum gradient support *Geophysics* **73** F45–54

Dahlin T, Leroux V and Nissen J 2002 Measuring techniques in induced polarisation imaging *J. Appl. Geophys.* **50** 279–98

Emerson D W 1969 Laboratory electrical resistivity measurements of rocks *Proc. Aust. Inst. Min. Met.* **230** 51–62

Eskola L 1992 *Geophysical Interpretation using Integral Equations* (London: Chapman and Hall)

Guptasarma D 1983 Effect of surface polarization on resistivity modeling *Geophysics* **48** 98–106

Hördt A, Blaschek R, Kemna A and Zisser N 2007 Hydraulic conductivity estimation from induced polarisation data at the field scale—the Krauthausen case history *J. Appl. Geophys.* **62** 33–46

Kemna A, Binley A, Ramirez A and Daily W 2000 Complex resistivity tomography for environmental applications *Chem. Eng. J.* **77** 11–8

Kemna A, Binley A and Slater L 2004 Crosshole IP imaging for engineering and environmental applications *Geophysics* **69** 97–107

Kemna A, Münch H M, Titov K, Zimmermann E and Vereecken H 2005 Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity *Proc. EAGE Near Surface* P054, 4 p

Leroy P, Revil A, Kemna A, Cosenza P and Ghorbani A 2008 Complex conductivity of water-saturated packs of glass beads *J. Colloid Interface Sci.* **321** 103–17

Lesmes D P and Frye K M 2001 Influence of pore fluid chemistry on the complex conductivity and induced polarization responses of Berea sandstone *J. Geophys. Res.* **106** 4079–90

Macdonald J R 1987 *Impedance Spectroscopy* (New York: Wiley)

Morrison R 1998 *Grounding and Shielding Techniques* (New York: Wiley)

Pelton W H, Ward S H, Hallof P G, Sill W R and Nelson P H 1978 Mineral discrimination and removal of inductive coupling with multi-frequency IP *Geophysics* **43** 588–609

Scott J B T and Barker R D 2003 Determining pore-throat size in Permo-Triassic sandstones from low frequency electrical spectroscopy *Geophys. Res. Lett.* **30** 1450 doi:10.1029/2003GL016951

Slater L and Binley A 2006 Synthetic and field-based electrical imaging of a zerovalent iron barrier: implications for monitoring long-term barrier performance *Geophysics* **71** B129–37

Slater L and Lesmes D 2002 Electrical–hydraulic relationships observed for unconsolidated sediments *Water Resources Res.* **38** 1213–25

Sumner J S 1976 *Principles of Induced Polarization for Geophysical Exploration* (Amsterdam: Elsevier)

Tietze U and Schenk C 1985 *Halbleiterschaltungstechnik* (Berlin: Springer)

Ulrich C and Slater L D 2004 Induced polarization measurements on unsaturated, unconsolidated sands *Geophysics* **69** 762–71

Vanhala H 1997 Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method *Geophys. Prospect.* **45** 303–26

Vanhala H and Soininen H 1995 Laboratory technique for measurement of spectral induced polarization response of soil samples *Geophys. Prospect.* **43** 655–76